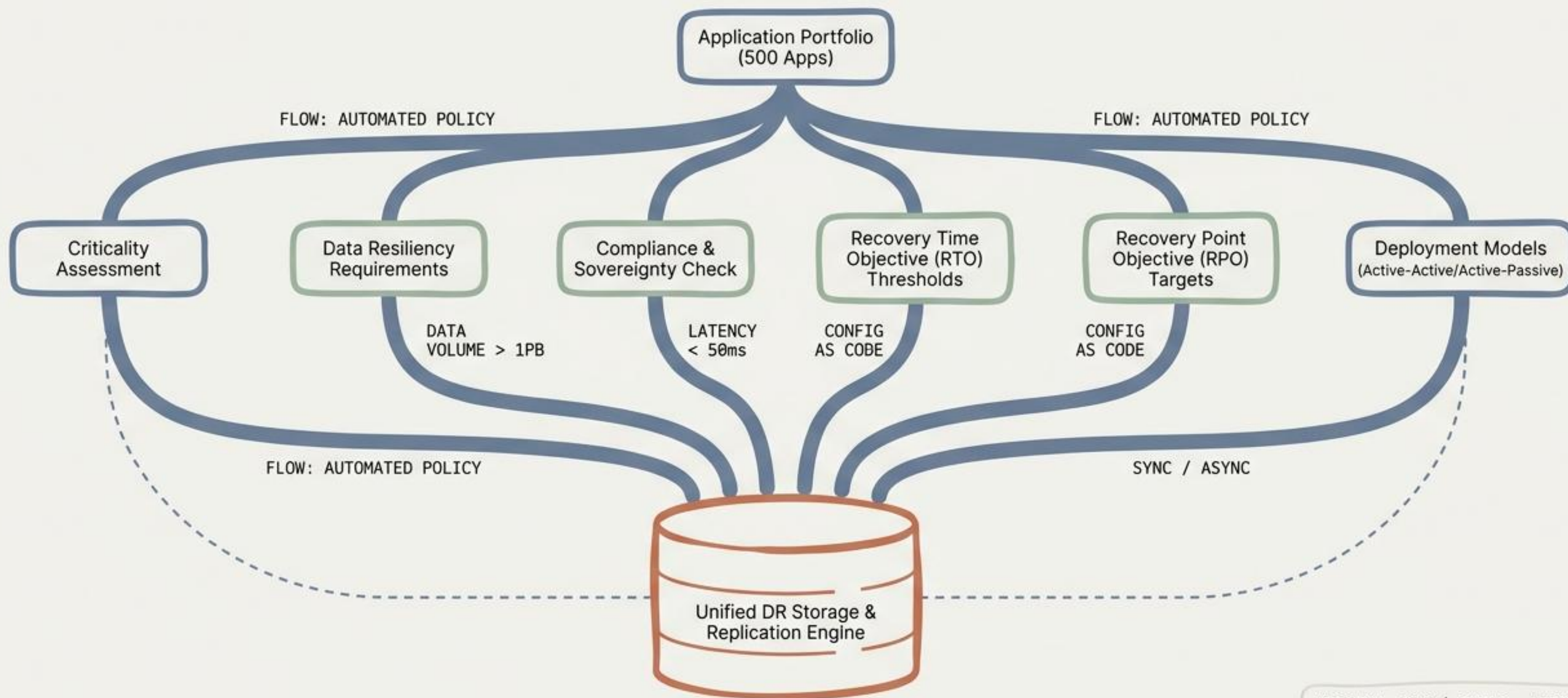
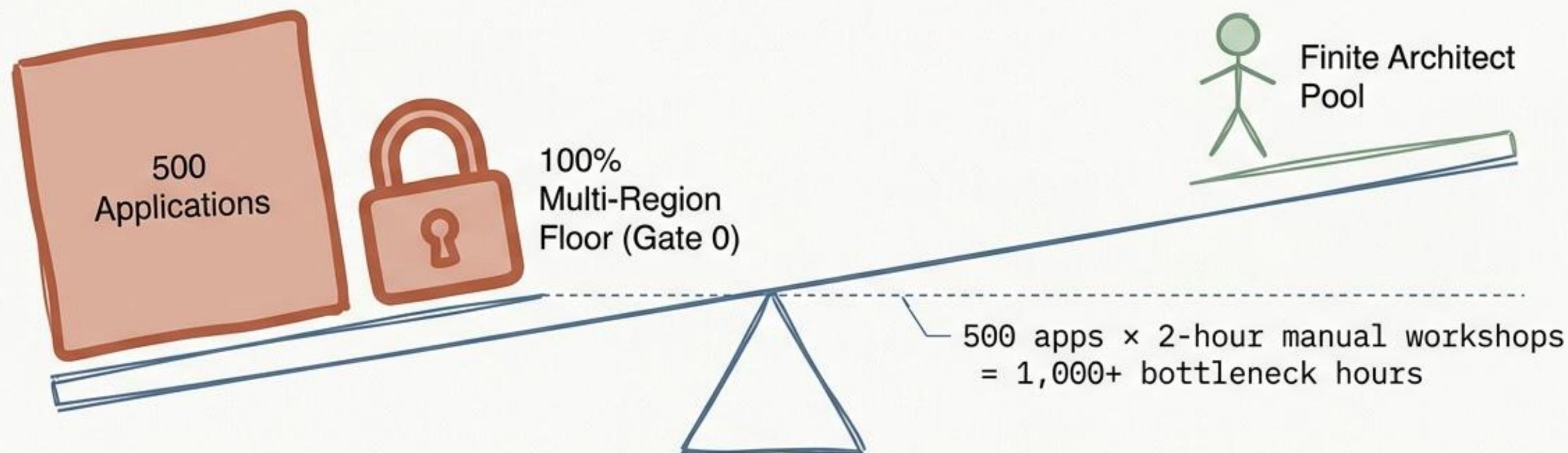


The Multi-Region DR Decision Engine

Scaling Architecture & Governance Across 500 Applications



The Mandate & The Math



The floor for all applications is now cross-region Backup & Restore. No app exits at Multi-AZ only. The question is no longer if, but how far above the floor each app sits. We require velocity and consistency that manual architecture workshops simply cannot provide.

Four Pillars of Automated Governance



Tree-as-Code

The 5-band DR tree is authored once as `dr-decision-tree.yaml` in Git. A single source of truth.



Server-Side Compute

Channels collect evidence; a central Python service computes the strategy. Client-side drift is impossible.



Pre-Seeded Context

RTO/RPO limits are pre-populated from the registry. Teams confirm data; they don't enter it from scratch.

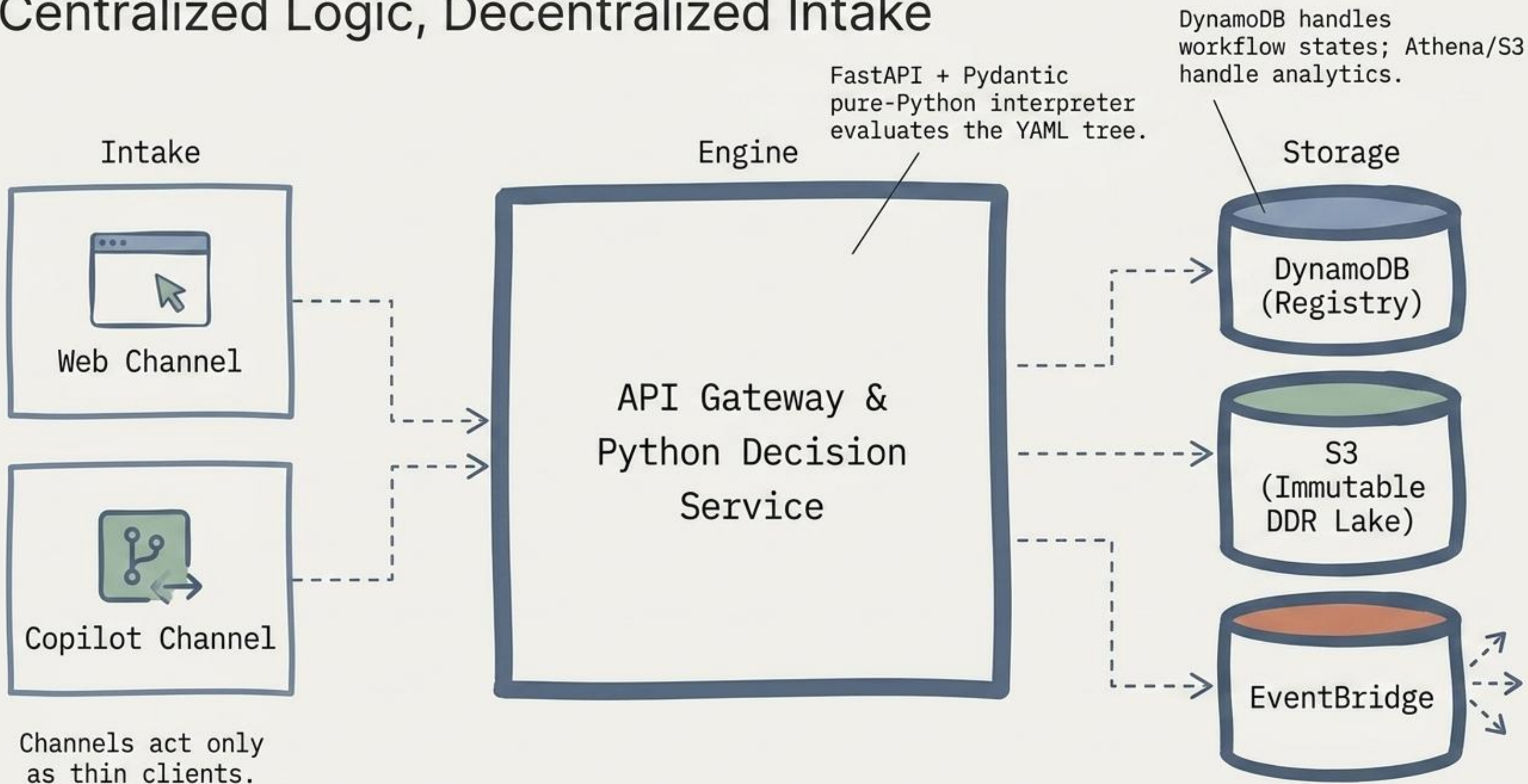


Immutable Records

Every submission generates an append-only, versioned DR Decision Record (DDR).

Shifting the workload from data gathering to confirmation and justification.

Centralized Logic, Decentralized Intake



Dual Intake Channels

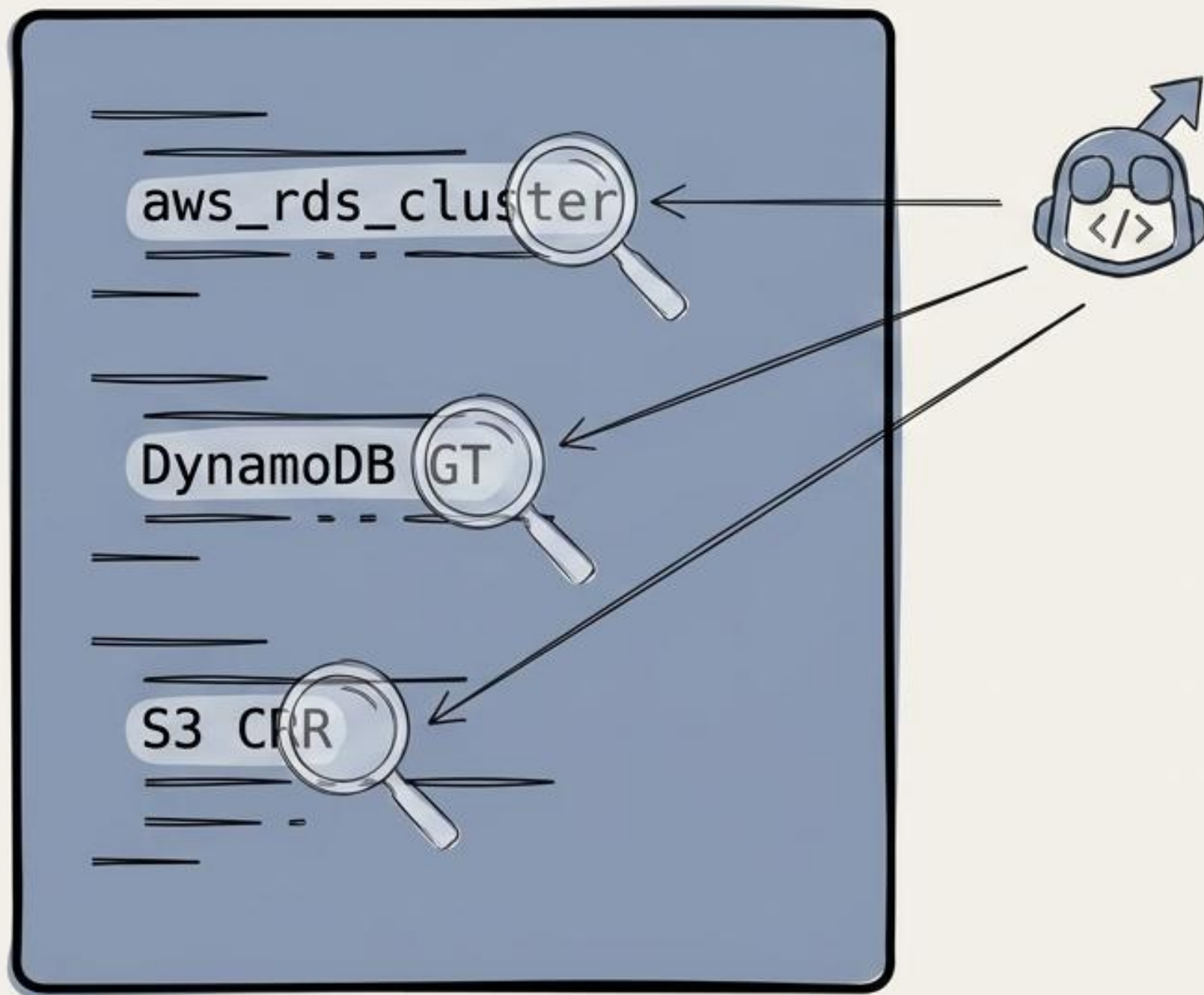
 Channel A (Forge Web Module)	 Channel B (GitHub Copilot Extension)
• Audience: App Owners & PMs • UX: Guided schema-driven UI Wizard • Tech Stack: React + TypeScript + tRPC	• Audience: Engineers & Developers • UX: Conversational AI & Claude Code plugin • Tech Stack: Agent Skills Standard



byte-identical ddr.json

Different workflows, identical outcomes. Both channels hit the same API and produce the exact same validated JSON schema.

Channel B's Unfair Advantage: Repo Introspection



- Instead of asking what the data layer is, Copilot scans Terraform, CDK, and Helm.
- It detects `global_cluster_identifier`, SaaS drivers, and VPC endpoints automatically.
- Generates evidence-backed answers for Gate 3, eliminating self-reported guesses.

Hard Rule — The skill contains no strategy logic. It extracts evidence and calls the central API.

The DR Decision Record (DDR)

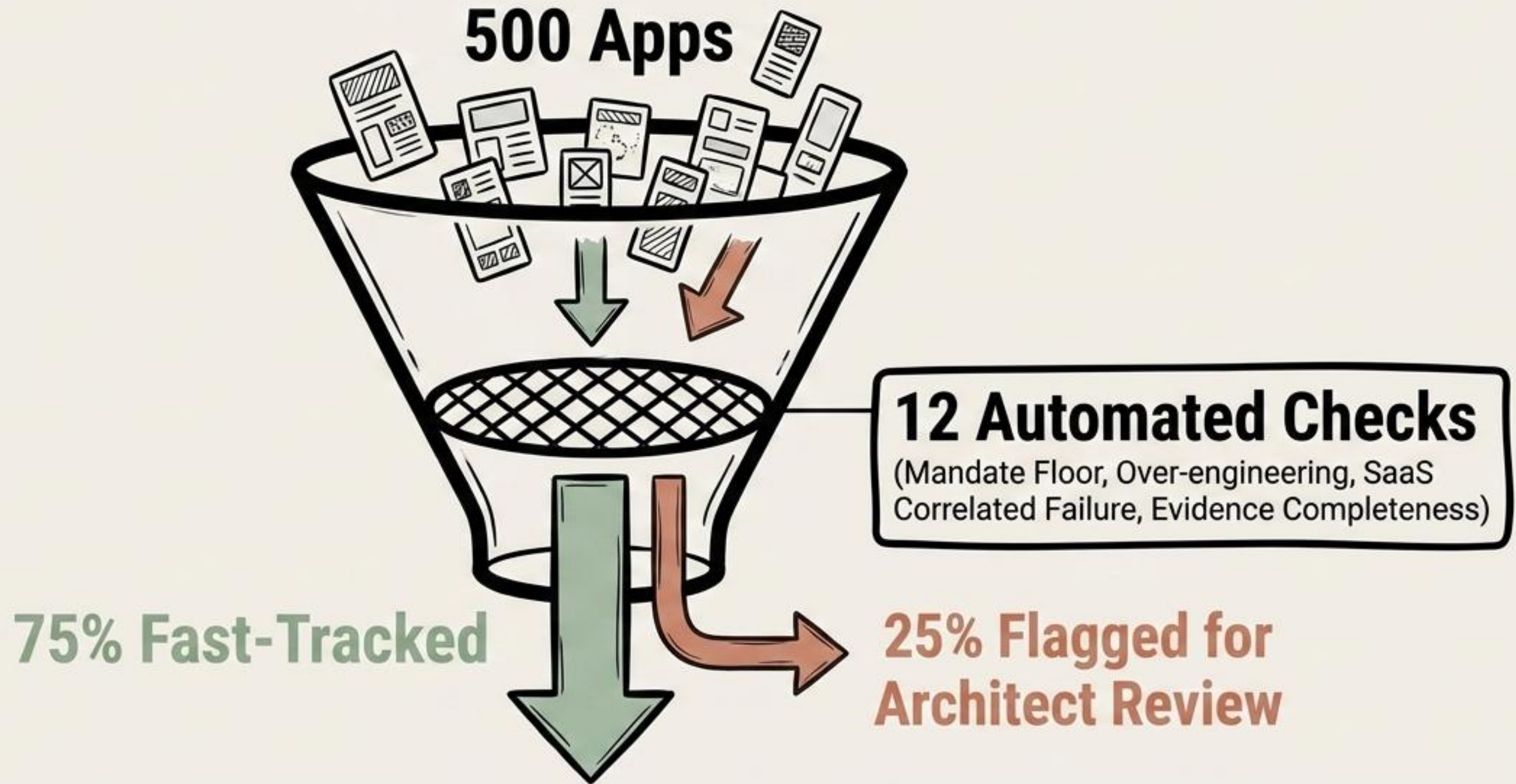
Modeled on Architecture Decision Records (ADRs).
Append-only, fully auditable.



The Analytical Heart:
The mathematical delta between
computed and declared.

Agreement = Fast-track approval.
Divergence = Trigger for deviation
workflow and CFO cost analysis.

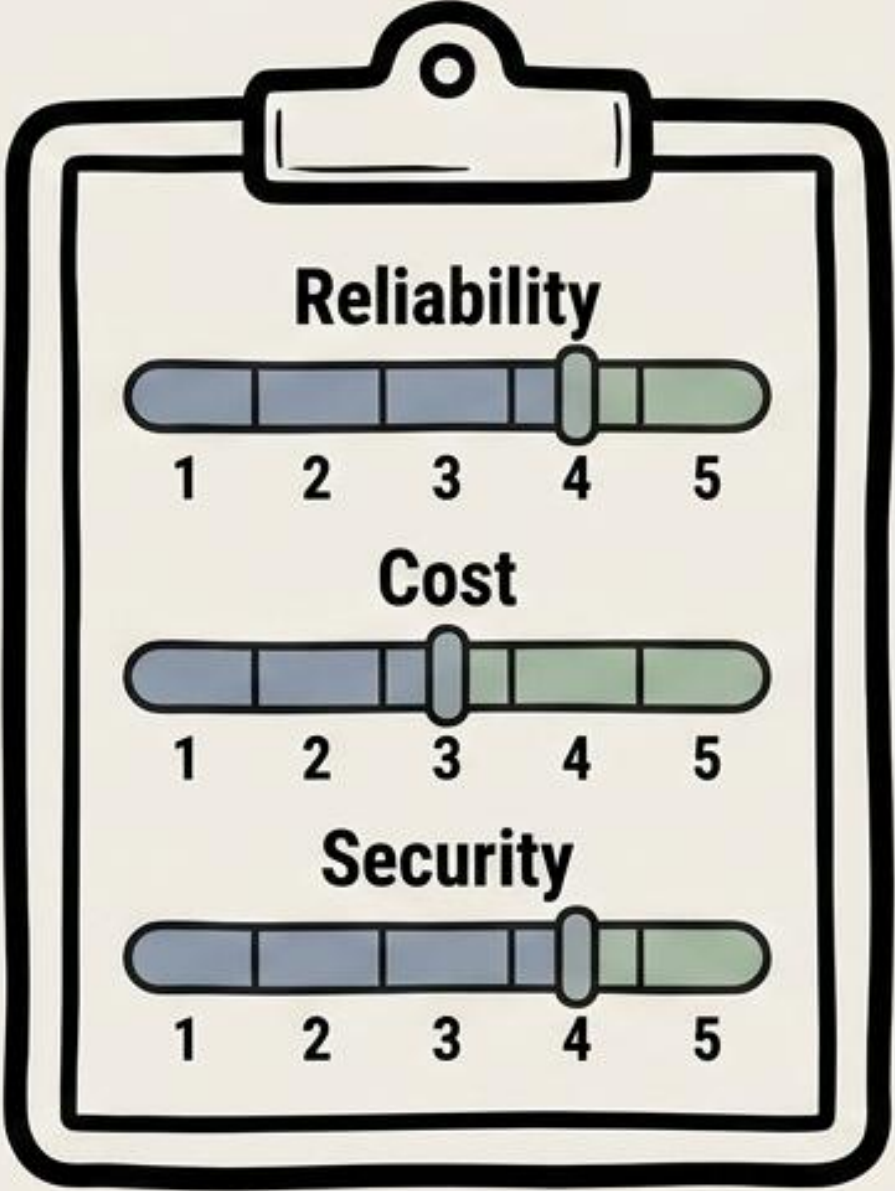
Layer 1: Automated Conformance



Clean submissions where computed == declared bypass human review. By instantly filtering out 75% of submissions, finite architect capacity is concentrated exclusively on high-risk deviations

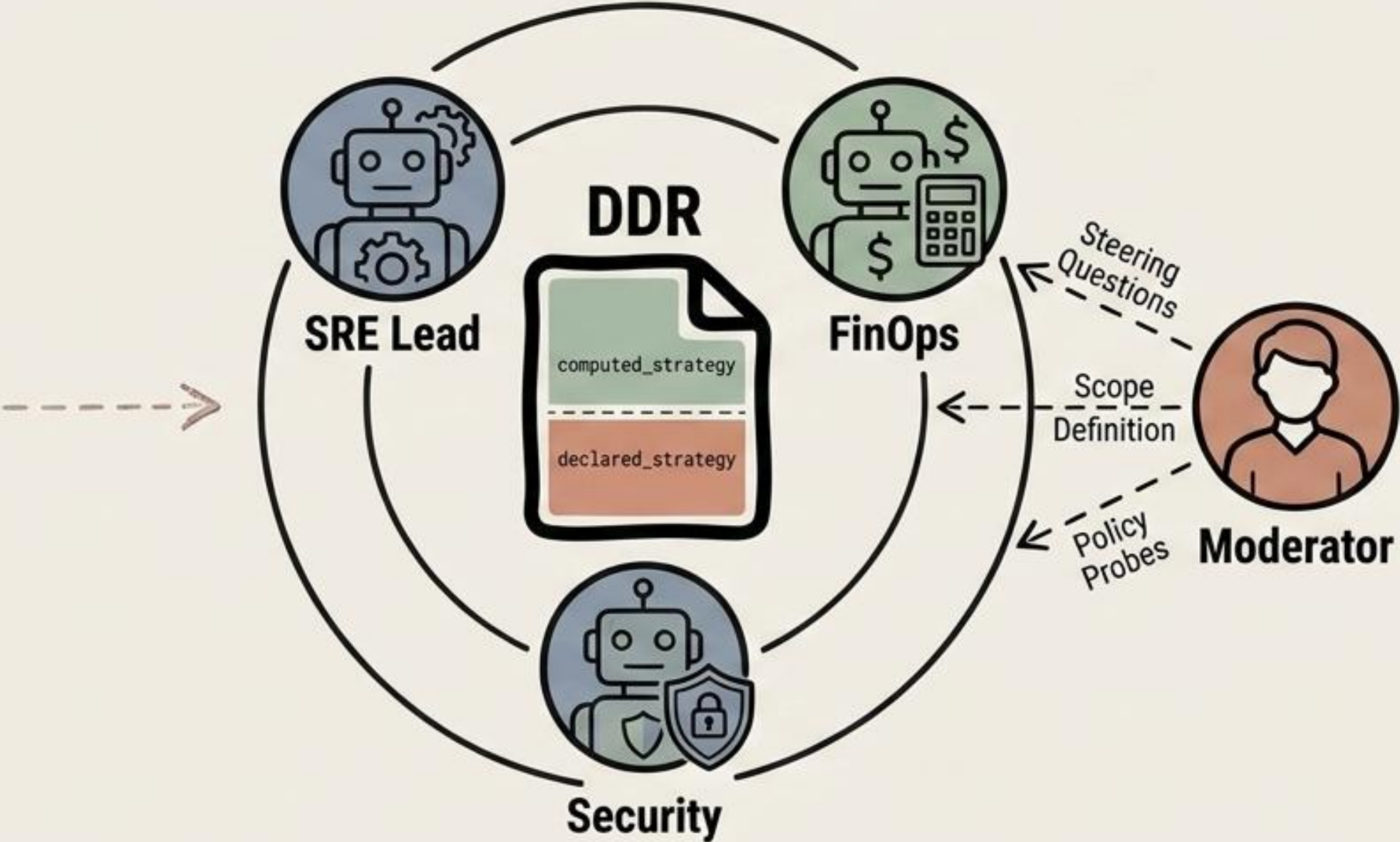
Layers 2 & 3: Rigorous Escalation

Human Review Rubric



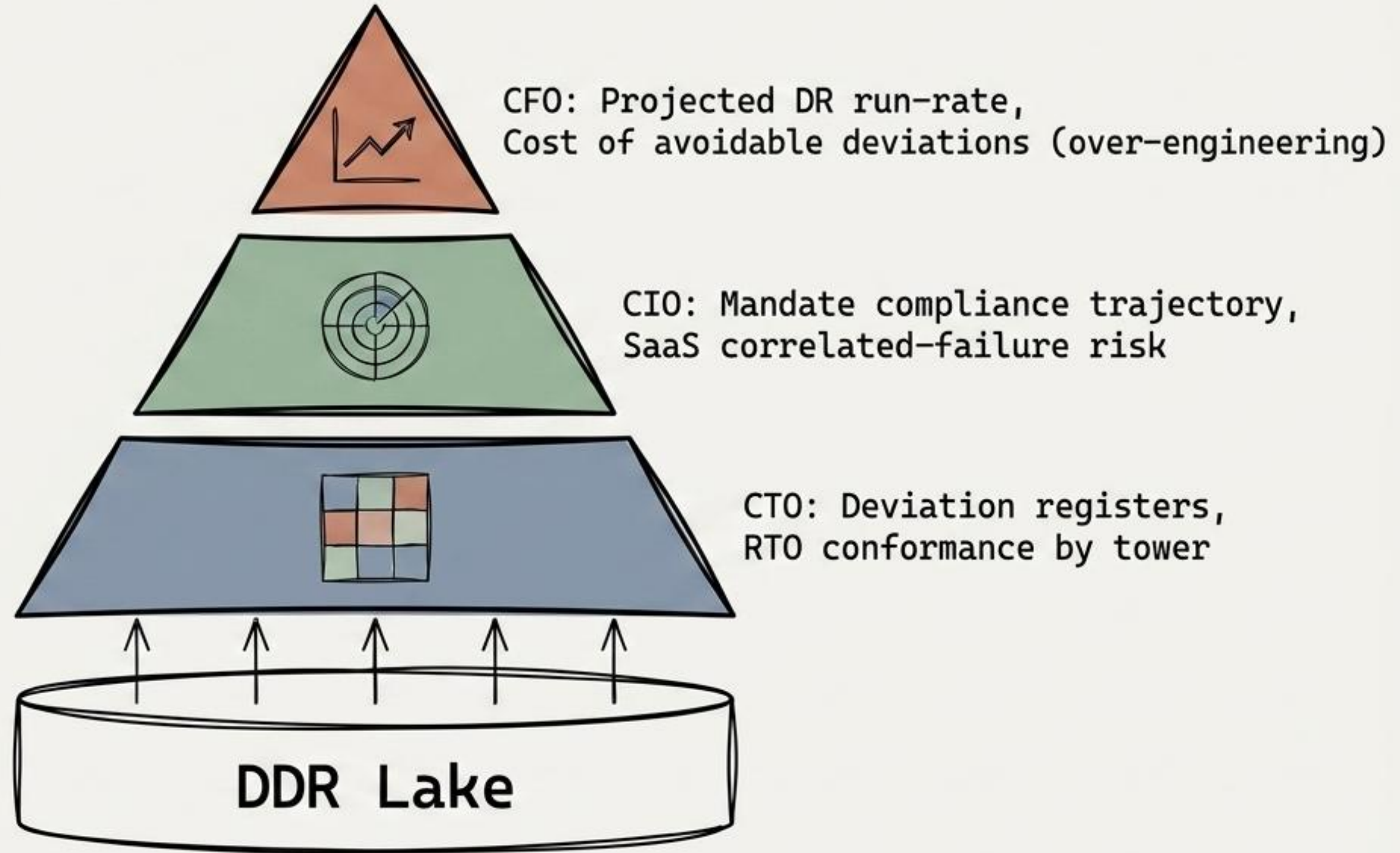
Layer 2: Flagged apps scored 1-5 against Well-Architected dimensions. High-risk or extreme deviations trigger Layer 3.

Co-STORM AI Challenge Panel



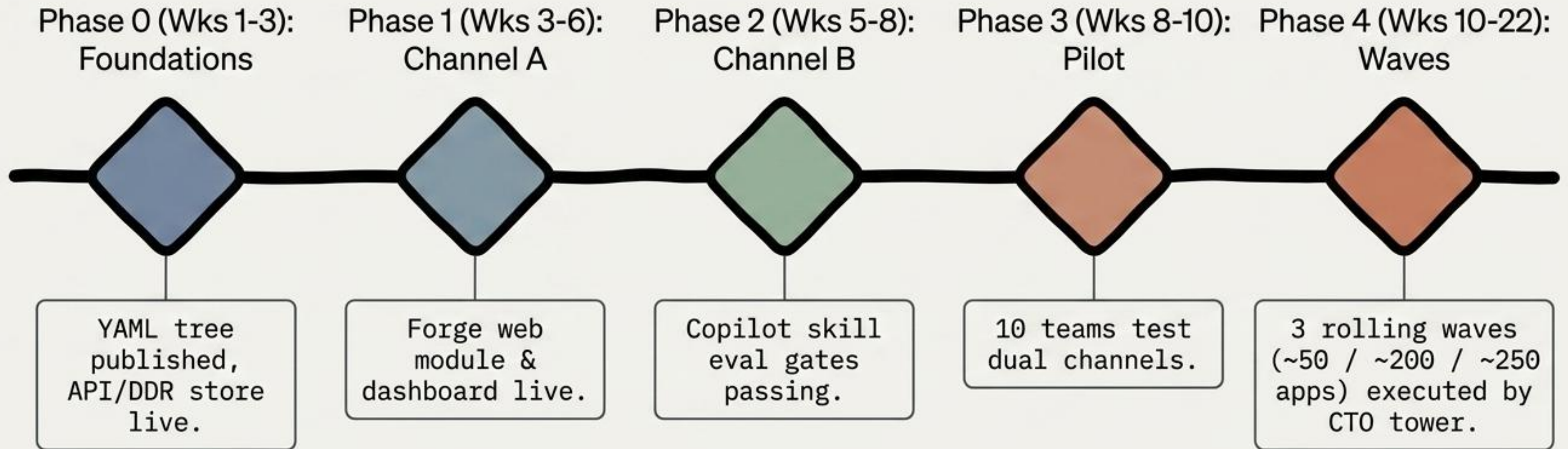
Layer 3: Expert AI agents interrogate the DDR to generate a cited verdict memo and issue **mind-map**, ensuring a consistent challenge process across all CTO  NotebookLM

Executive Reporting Cascade



One centralized dataset, three distinct lenses. Zero manual slide decks.
All readouts are generated live from the DDR single-source-of-truth.

The Delivery Masterplan



TARGET: <30 MIN MEDIAN TIME | ≥75% FAST-TRACK | ZERO SCHEMA DRIFT

Defensive Architecture: Built for Reality



Hazard: Teams game the wizard.

Shield: Evidence-required gates & 10% audit sampling.



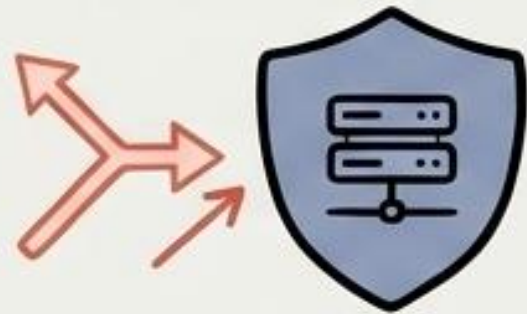
Hazard: Data-layer answers are blind guesses.

Shield: Copilot directly extracts IaC configs.



Hazard: Architect review bottleneck.

Shield: 12 automated checks clear 75% of traffic.



Hazard: Dual channels diverge.

Shield: Pure client/server separation; strict GitHub eval gates.

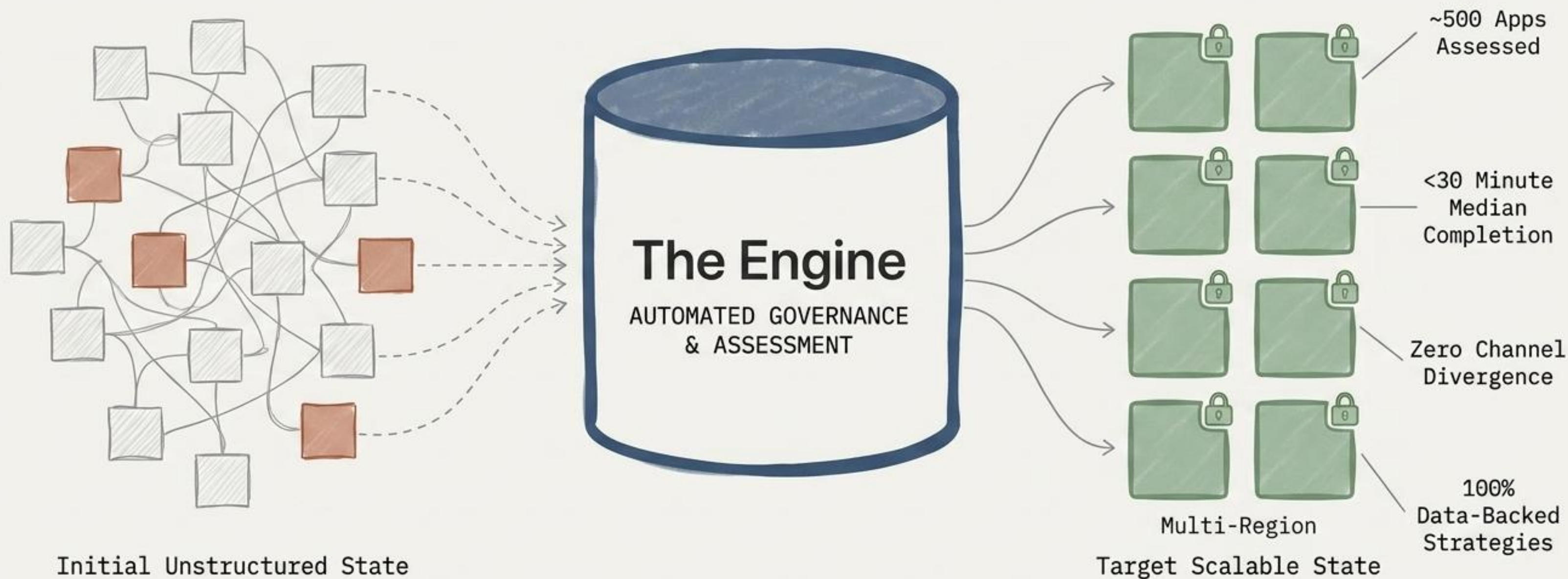


Hazard: Vendor DR claims are exaggerated.

Shield: Central Vendor DR Catalog enforces configured vs capable.

The system is engineered to absorb human nature, scaling frictionlessly while maintaining cryptographic-level data integrity.

The Target State: Scalable Resilience



Moving enterprise resilience from manual, untracked workshops to a highly scalable, automated governance engine.